

Binary Logistic Regression: Statistical & Machine Learning Approaches

Comprehensive Training Notebook

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Abstract

This presentation provides a comprehensive guide to Binary Logistic Regression, transitioning from data preprocessing to both predictive machine learning and inferential statistical modeling.

- **Foundations:** The mathematical theory behind the sigmoid function and log-odds.
- **Data Engineering:** Handling missing values, one-hot encoding, and avoiding the dummy variable trap.
- **Machine Learning:** Training a Scikit-Learn model, evaluating via Accuracy, Confusion Matrix, and ROC-AUC.
- **Statistical Inference:** Using Statsmodels to extract p-values, confidence intervals, and clinically interpretable Odds Ratios.

We emphasize the translation of statistical outputs into actionable, clinically sound research reporting using the Heart Disease dataset.

Session Objectives

- Understand the mathematical foundation of Binary Logistic Regression (Sigmoid & Logit).
- Preprocess clinical data: handle missing values and encode categorical variables correctly.
- Implement and evaluate a predictive model using Scikit-Learn (Accuracy, ROC-AUC).
- Fit a statistical Logit model using Statsmodels to assess predictor significance.
- Calculate and interpret Odds Ratios (OR) for clinical decision-making.
- Distinguish between the Machine Learning approach (prediction) and the Statistical approach (inference).

Part 1: Foundations of Binary Logistic Regression

"Modeling probabilities and the log-odds transformation."

What is Binary Logistic Regression?

- Used when the dependent variable (target) is **binary** (e.g., 0 or 1, Yes or No, Disease or Healthy).
- Unlike Linear Regression, which predicts continuous values, Logistic Regression predicts the **probability** of an event occurring.
- The output is strictly bounded between 0 and 1.

The Core Concept

We model the probability $P(Y = 1|X)$ using the **Logistic (Sigmoid) Function**, which maps any real-valued number into the $(0, 1)$ interval.

The Logistic Function & Log-Odds

The Sigmoid Function

$$P(Y = 1) = \frac{1}{1 + e^{-z}} \quad \text{where} \quad z = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$$

- To make the relationship linear, we apply the **Logit (Log-Odds)** transformation:

$$\text{Logit}(P) = \ln \left(\frac{P}{1 - P} \right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$$

- **Odds:** The ratio of the probability of the event happening to it not happening $\left(\frac{P}{1 - P} \right)$.
- **Coefficients (β):** Represent the change in the *log-odds* for a 1-unit increase in the predictor.

Part 2: Dataset & Preprocessing

"Preparing clinical data for modeling."

The Heart Disease Dataset

- **Source:** Heart.csv containing clinical data for 303 patients.
- **Target Variable:** HD (Atherosclerotic Heart Disease).
 - **0:** No Heart Disease
 - **1:** Presence of Heart Disease
- **Features:** 13 demographic, clinical, and diagnostic variables.
- **Note:** The redundant string column AHD was dropped to prevent data leakage.

Variable Definitions & Clinical Relevance

Variable	Definition	Clinical Relevance
Age	Patient's age (years)	Strong predictor of cardiovascular risk
Sex	Biological sex (0=F, 1=M)	Men have higher baseline CVD risk
ChestPain	Type of pain experienced	Asymptomatic often indicates severe disease
RestBP	Resting blood pressure	Hypertension is a major CVD risk factor
MaxHR	Maximum heart rate achieved	Lower values indicate poor cardiac function
Oldpeak	ST depression induced by exercise	Marker of myocardial ischemia
Ca	Number of major vessels (0-3)	More vessels = more severe disease

Handling Missing Values

- The dataset contained missing values encoded as 'NA' strings, read as NaN by pandas.
- Ca (**Number of major vessels**): 4 missing values.
 - **Action:** Filled with the **median** (robust to outliers).
- Tha1 (**Thalassemia stress test**): 2 missing values.
 - **Action:** Filled with the **mode** (most frequent category).

Result

A complete dataset of 303 entries with 0 null values, ensuring no data loss during model training.

Categorical Encoding & The Dummy Variable Trap

- Variables like `ChestPain` and `Thal` are nominal and must be converted to numbers.
- We use **One-Hot Encoding** (`pd.get_dummies`).
- **The Dummy Variable Trap:** If a category has K levels, creating K dummy variables causes perfect multicollinearity (they sum to 1, matching the intercept).
- **Solution:** Use `drop_first=True` to omit one category per variable, serving as the baseline reference.

Result

The feature matrix expanded to **16 columns** (11 numeric + 5 dummy variables).

Part 3: Exploratory Data Analysis (EDA)

"Visualizing relationships and distributions."

EDA: Correlation & Target Distribution

- **Correlation Heatmap:** Examines pairwise linear relationships between numeric features.
 - Helps identify potential multicollinearity before modeling.
- **Target Distribution:** Visualizes the class balance of Heart Disease (0 vs 1).
 - Ensures the dataset isn't severely imbalanced, which could bias the model.

Insight

EDA confirms that variables like MaxHR and Oldpeak have strong correlations with the target, making them vital predictors.

EDA: Age Distribution & Heart Rate vs Oldpeak

- **Age Distribution:** A stacked histogram shows that patients with heart disease tend to be older, shifting the distribution to the right.
- **MaxHR vs Oldpeak:** A scatter plot reveals a clear pattern:
 - Patients with heart disease generally exhibit **lower maximum heart rates** and **higher ST depression (Oldpeak)**.

Clinical Takeaway

These visualizations align with medical literature: exercise-induced ischemia (high Oldpeak) and poor chronotropic competence (low MaxHR) are strong indicators of cardiovascular disease.

Part 4: Data Partitioning & Scaling

"Setting up the data for robust modeling."

Train-Test Split & Stratification

- **Split Ratio:** 80% Training (242 samples), 20% Testing (61 samples).
- **Random State:** Set to 42 for reproducibility.
- **Stratification** (`stratify=y`):
 - Ensures the proportion of heart disease cases (0s and 1s) is maintained identically in both the train and test sets.
 - Prevents the model from being trained or evaluated on a skewed distribution.

Feature Scaling (Standardization)

- Applied `StandardScaler` to normalize features (mean = 0, variance = 1).
- **Why is this crucial?**
 - ① **Convergence:** Speeds up the gradient descent optimization algorithm used by Scikit-Learn.
 - ② **Coefficient Comparison:** Allows us to compare the magnitude of coefficients directly to determine feature importance.
 - ③ **Regularization:** Ensures penalties (L1/L2) are applied fairly across all features.

Note

The scaler is *fit* only on the training data to prevent data leakage, then *transformed* on the test data.

Part 5: Machine Learning Approach (Scikit-Learn)

"Maximizing predictive accuracy."

Training the Scikit-Learn Model

- **Algorithm:** LogisticRegression from `sklearn.linear_model`.
- **Parameters:**
 - `max_iter=1000`: Ensures the solver has enough iterations to converge.
 - `random_state=42`: For reproducible results.
- **Focus:** The ML approach prioritizes **predictive performance** on unseen data over statistical inference.

Model Output

The model successfully converged in 12 iterations, learning the weights for all 16 features.

ML Evaluation: Accuracy & Classification Report

Class	Precision	Recall	F1-Score	Support
0 (No HD)	0.94	0.88	0.91	33
1 (Yes HD)	0.87	0.93	0.90	28
Accuracy			0.9016	61

- **Accuracy (90.16%):** The model correctly classified 90% of the patients.
- **Precision/Recall:** Well-balanced. The model is slightly better at catching actual heart disease cases (Recall = 0.93) without generating too many false alarms (Precision = 0.87).

ML Evaluation: Confusion Matrix

- Visualizes the four possible outcomes of a binary classifier:
 - **True Negatives (TN)**: Correctly predicted healthy.
 - **False Positives (FP)**: Healthy patient incorrectly flagged as sick (Type I error).
 - **False Negatives (FN)**: Sick patient incorrectly flagged as healthy (Type II error).
 - **True Positives (TP)**: Correctly predicted heart disease.

Clinical Context

In medical diagnostics, minimizing **False Negatives** is often critical. Our model's high recall (0.93) indicates it successfully minimizes missed diagnoses.

ML Evaluation: ROC Curve & AUC

- **ROC Curve:** Plots the True Positive Rate (Sensitivity) against the False Positive Rate (1 - Specificity) across all classification thresholds.
- **AUC (Area Under the Curve):** A single scalar value summarizing the model's discriminative ability.

Result

ROC-AUC Score: 0.9545

An AUC of 0.95 is considered **excellent**. It means there is a 95.45% probability that the model will rank a randomly chosen positive patient higher than a randomly chosen negative patient.

Part 6: Statistical Approach (Statsmodels)

"Understanding relationships and statistical significance."

Fitting the Logit Model (Statsmodels)

- While Scikit-Learn focuses on prediction, statsmodels focuses on **statistical inference**.
- **Model:** `sm.Logit(y_train, X_train_sm)`.
- **Constant:** We must explicitly add a constant (`sm.add_constant`) to estimate the intercept (β_0).
- **Method:** Maximum Likelihood Estimation (MLE).

Goal

To determine *which* variables significantly affect the odds of heart disease and by *how much*.

Statistical Model Summary

Predictor	coef	std err	z	P>—z—
Sex	1.3469	0.510	2.642	0.008 **
MaxHR	-0.0272	0.010	-2.829	0.005 **
Ca	1.3585	0.318	4.271	0.000 ***
ChestPain_nonanginal	-1.5525	0.532	-2.916	0.004 **
ChestPain_typical	-1.8377	0.693	-2.651	0.008 **

- **Pseudo R-squared:** 0.4965 (Explains 50% of the variance in the log-odds).
- **Significant Predictors ($p < 0.05$):** Sex, MaxHR, Ca, and specific ChestPain types.

Interpreting Statistical Coefficients

- Coefficients represent the change in the **log-odds** for a 1-unit increase in the predictor.
- **Positive Coefficient:** Increases the log-odds (and probability) of the event.
 - Ca (1.3585): More major vessels strongly increases the log-odds of heart disease.
- **Negative Coefficient:** Decreases the log-odds.
 - MaxHR (-0.0272): Higher max heart rate decreases the log-odds of heart disease.

Limitation

Log-odds are not intuitively understandable for clinical reporting. We must convert them to **Odds Ratios**.

Part 7: Odds Ratios & Interpretation

"Translating math into clinical insights."

Understanding Odds Ratios

Definition

The Odds Ratio (OR) is the exponentiated coefficient: $OR = e^{\beta}$.

- It represents the **multiplicative change** in the **odds** of the outcome for a 1-unit increase in the predictor, holding all other variables constant.
- **OR > 1**: Increases the odds of the event.
- **OR < 1**: Decreases the odds of the event (Protective factor).
- **OR = 1**: No effect on the odds.

Calculating Odds Ratios

Predictor	2.5% CI	97.5% CI	Odds Ratio (e^{β})
Age	0.927	1.017	0.971
Sex	1.415	10.444	3.845
MaxHR	0.955	0.991	0.973
Ca	2.085	7.256	3.890
ChestPain_typical	0.040	0.619	0.159

- The Confidence Intervals (CI) for Sex, MaxHR, and Ca **do not cross 1**, confirming their statistical significance.

Interpreting Key Odds Ratios

Ca (Odds Ratio = 3.890)

For each additional major vessel colored by fluoroscopy, the **odds of heart disease increase by 289%** (or are multiplied by 3.89), holding all other factors constant.

ChestPain_typical (Odds Ratio = 0.159)

Patients experiencing typical chest pain have **84.1% lower odds** of heart disease compared to the baseline category (asymptomatic).

Note: Asymptomatic pain often indicates more severe, advanced disease than typical angina.

Clinical Interpretation of Odds Ratios

- **Sex (OR = 3.845):** Males have nearly 4 times the odds of heart disease compared to females, confirming the known baseline gender disparity in cardiovascular risk.
- **MaxHR (OR = 0.973):** For every 1 bpm increase in maximum heart rate, the odds of heart disease decrease by 2.7%. A lower MaxHR is a strong indicator of poor cardiac function.

Actionable Insight

These Odds Ratios allow doctors to quantify risk. If a patient presents with 2 calcified vessels ($Ca=2$), their baseline odds are multiplied by $3.89^2 \approx 15$, signaling an urgent need for intervention.

Part 8: ML vs Statistical Approach

"Choosing the right tool for the job."

Scikit-Learn vs Statsmodels

Feature	Scikit-Learn (ML)	Statsmodels (Statistical)
Primary Goal	Prediction / Classification	Inference / Hypothesis Testing
Output	Accuracy, ROC-AUC, Classes	P-values, Confidence Intervals, ORs
Regularization	Built-in (L1, L2, ElasticNet)	Not default (requires manual setup)
Intercept	Handled automatically	Must add constant manually
Scaling	Crucial for convergence	Helpful for interpretation

When to use which approach?

- **Use Machine Learning (Scikit-Learn) when:**

- You need to deploy a model to predict outcomes for new patients.
- You have a large dataset and want to use regularization to prevent overfitting.
- Predictive accuracy is the sole metric of success.

- **Use Statistical Modeling (Statsmodels) when:**

- You need to understand the *relationship* between variables.
- You are writing a research paper or clinical report requiring p-values and Odds Ratios.
- You need to test specific hypotheses about individual predictors.

Part 9: Assumptions & Diagnostics

"Ensuring the validity of our models."

Assumptions of Logistic Regression

- ➊ **Binary Outcome:** The dependent variable must be binary.
- ➋ **Independence:** Observations must be independent (no repeated measures).
- ➌ **Linearity of Log-Odds:** Continuous predictors must have a linear relationship with the *log-odds* of the target.
- ➍ **No Multicollinearity:** Predictors should not be highly correlated with each other.
- ➎ **Large Sample Size:** Requires sufficient data (rule of thumb: at least 10 events per predictor variable).

Multicollinearity & Model Stability

- Occurs when independent variables are highly correlated.
- **Consequence:** Inflates standard errors, making coefficients unstable and p-values unreliable.
- **Detection:** Variance Inflation Factor (VIF).
- **Prevention in our workflow:**
 - Used `drop_first=True` during One-Hot Encoding to avoid the dummy variable trap.
 - Checked the correlation heatmap during EDA.

Part 10: Synthesis & Best Practices

"Bridging the gap between code and clinical decisions."

Best Practices for Logistic Regression Reporting

- 1 **Visualize First:** Always use EDA to understand distributions and correlations before modeling.
- 2 **Handle Data Rigorously:** Impute missing values logically (median/mode) and encode categoricals correctly.
- 3 **Scale for ML, Interpret for Stats:** Standardize data for Scikit-Learn, but remember scaling changes the meaning of statistical coefficients.
- 4 **Report Comprehensively:** Include Accuracy, AUC, Pseudo R^2 , and Confidence Intervals.
- 5 **Translate to Domain Language:** Never report raw log-odds to stakeholders. Always convert to Odds Ratios and explain the clinical impact.

Questions?

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